

Partial Derivatives

Now that we have the brief discussion on limits out of the way we can proceed into taking derivatives of functions of more than one variable. Before we actually start taking derivatives of functions of more than one variable let's recall an important interpretation of derivatives of functions of one variable.

Recall that given a function of one variable, $f(x)$, the derivative, $f'(x)$, represents the rate of change of the function as x changes. This is an important interpretation of derivatives and we are not going to want to lose it with functions of more than one variable. The problem with functions of more than one variable is that there is more than one variable. In other words, what do we do if we only want one of the variables to change, or if we want more than one of them to change? In fact, if we're going to allow more than one of the variables to change there are then going to be an infinite amount of ways for them to change. For instance, one variable could be changing faster than the other variable(s) in the function. Notice as well that it will be completely possible for the function to be changing differently depending on how we allow one or more of the variables to change.

We will need to develop ways, and notations, for dealing with all of these cases. In this section we are going to concentrate exclusively on only changing one of the variables at a time, while the remaining variable(s) are held fixed. We will deal with allowing multiple variables to change in a later [section](#).

Because we are going to only allow one of the variables to change taking the derivative will now become a fairly simple process. Let's start off this discussion with a fairly simple function.

Let's start with the function $f(x, y) = 2x^2y^3$ and let's determine the rate at which the function is changing at a point, (a, b) , if we hold y fixed and allow x to vary and if we hold x fixed and allow y to vary.

We'll start by looking at the case of holding y fixed and allowing x to vary. Since we are interested in the rate of change of the function at (a, b) and are holding y fixed this means that we are going to always have $y = b$ (if we didn't have this then eventually y would have to change in order to get to the point...). Doing this will give us a function involving only x 's and we can define a new function as follows,

$$g(x) = f(x, b) = 2x^2b^3$$

Now, this is a function of a single variable and at this point all that we are asking is to determine the rate of change of $g(x)$ at $x = a$. In other words, we want to compute $g'(a)$ and since this is a function of a single variable we already know how to do that. Here is the rate of change of the function at (a, b) if we hold y fixed and allow x to vary.

$$g'(a) = 4ab^3$$

We will call $g'(a)$ the **partial derivative** of $f(x, y)$ with respect to x at (a, b) and we will denote it in the following way,

$$f_x(a, b) = 4ab^3$$

Now, let's do it the other way. We will now hold x fixed and allow y to vary. We can do this in a similar way. Since we are holding x fixed it must be fixed at $x = a$ and so we can define a new function of y and then differentiate this as we've always done with functions of one variable.

Here is the work for this,

$$h(y) = f(a, y) = 2a^2y^3 \quad \Rightarrow \quad h'(b) = 6a^2b^2$$

In this case we call $h'(b)$ the **partial derivative** of $f(x, y)$ with respect to y at (a, b) and we denote it as follows,

$$f_y(a, b) = 6a^2b^2$$

Note that these two partial derivatives are sometimes called the **first order partial derivatives**. Just as with functions of one variable we can have derivatives of all orders. We will be looking at higher order derivatives in a later [section](#).

Note that the notation for partial derivatives is different than that for derivatives of functions of a single variable. With functions of a single variable we could denote the derivative with a single prime. However, with partial derivatives we will always need to remember the variable that we are differentiating with respect to and so we will subscript the variable that we differentiated with respect to. We will shortly be seeing some alternate notation for partial derivatives as well.

Note as well that we usually don't use the (a, b) notation for partial derivatives as that implies we are working with a specific point which we usually are not doing. The more standard notation is to just continue to use (x, y) . So, the partial derivatives from above will more commonly be written as,

$$f_x(x, y) = 4xy^3 \quad \text{and} \quad f_y(x, y) = 6x^2y^2$$

Now, as this quick example has shown taking derivatives of functions of more than one variable is done in pretty much the same manner as taking derivatives of a single variable. To compute $f_x(x, y)$ all we need to do is treat all the y 's as constants (or numbers) and then differentiate the x 's as we've always done. Likewise, to compute $f_y(x, y)$ we will treat all the x 's as constants and then differentiate the y 's as we are used to doing.

Before we work any examples let's get the formal definition of the partial derivative out of the way as well as some alternate notation.

Since we can think of the two partial derivatives above as derivatives of single variable functions it shouldn't be too surprising that the definition of each is very similar to the definition of the derivative for single variable functions. Here are the formal definitions of the two partial derivatives we looked at above.

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \qquad f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}$$

If you recall the Calculus I definition of the limit these should look familiar as they are very close to the Calculus I definition with a (possibly) obvious change.

Now let's take a quick look at some of the possible alternate notations for partial derivatives. Given the function $z = f(x, y)$ the following are all equivalent notations,

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(f(x, y)) = z_x = \frac{\partial z}{\partial x} = D_x f$$

$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(f(x, y)) = z_y = \frac{\partial z}{\partial y} = D_y f$$

For the fractional notation for the partial derivative notice the difference between the partial derivative and the ordinary derivative from single variable calculus.

$$f(x) \quad \Rightarrow \quad f'(x) = \frac{df}{dx}$$

$$f(x, y) \quad \Rightarrow \quad f_x(x, y) = \frac{\partial f}{\partial x} \quad \& \quad f_y(x, y) = \frac{\partial f}{\partial y}$$

Okay, now let's work some examples. When working these examples always keep in mind that we need to pay very close attention to which variable we are differentiating with respect to. This is important because we are going to treat all other variables as constants and then proceed with the derivative as if it was a function of a single variable. If you can remember this you'll find that doing partial derivatives are not much more difficult than doing derivatives of functions of a single variable as we did in Calculus I.

Example 1 Find all of the first order partial derivatives for the following functions.

(a) $f(x, y) = x^4 + 6\sqrt{y} - 10$

(b) $w = x^2y - 10y^2z^3 + 43x - 7 \tan(4y)$

(c) $h(s, t) = t^7 \ln(s^2) + \frac{9}{t^3} - \sqrt[7]{s^4}$

(d) $f(x, y) = \cos\left(\frac{4}{x}\right) e^{x^2y - 5y^3}$

Solution

(a) $f(x, y) = x^4 + 6\sqrt{y} - 10$

Let's first take the derivative with respect to x and remember that as we do so all the y 's will be treated as constants. The partial derivative with respect to x is,

$$f_x(x, y) = 4x^3$$

Notice that the second and the third term differentiate to zero in this case. It should be clear why the third term differentiated to zero. It's a constant and we know that constants always differentiate to zero. This is also the reason that the second term differentiated to zero. Remember that since we are differentiating with respect to x here we are going to treat all y 's as constants. That means that terms that only involve y 's will be treated as constants and hence will differentiate to zero.

Now, let's take the derivative with respect to y . In this case we treat all x 's as constants and so the first term involves only x 's and so will differentiate to zero, just as the third term will. Here is the partial derivative with respect to y .

$$f_y(x, y) = \frac{3}{\sqrt{y}}$$

(b) $w = x^2y - 10y^2z^3 + 43x - 7 \tan(4y)$

With this function we've got three first order derivatives to compute. Let's do the partial derivative with respect to x first. Since we are differentiating with respect to x we will treat all y 's and all z 's as constants. This means that the second and fourth terms will differentiate to zero since they only involve y 's and z 's.

This first term contains both x 's and y 's and so when we differentiate with respect to x the y will be thought of as a multiplicative constant and so the first term will be differentiated just as the third term will be differentiated.

Here is the partial derivative with respect to x .

$$\frac{\partial w}{\partial x} = 2xy + 43$$

Let's now differentiate with respect to y . In this case all x 's and z 's will be treated as constants. This means the third term will differentiate to zero since it contains only x 's while the x 's in the first term and the z 's in the second term will be treated as multiplicative constants. Here is the derivative with respect to y .

$$\frac{\partial w}{\partial y} = x^2 - 20yz^3 - 28\sec^2(4y)$$

Finally, let's get the derivative with respect to z . Since only one of the terms involve z 's this will be the only non-zero term in the derivative. Also, the y 's in that term will be treated as multiplicative constants. Here is the derivative with respect to z .

$$\frac{\partial w}{\partial z} = -30y^2z^2$$

(c) $h(s, t) = t^7 \ln(s^2) + \frac{9}{t^3} - \sqrt[7]{s^4}$

With this one we'll not put in the detail of the first two. Before taking the derivative let's rewrite the function a little to help us with the differentiation process.

$$h(s, t) = t^7 \ln(s^2) + 9t^{-3} - s^{\frac{4}{7}}$$

Now, the fact that we're using s and t here instead of the "standard" x and y shouldn't be a problem. It will work the same way. Here are the two derivatives for this function.

$$h_s(s, t) = \frac{\partial h}{\partial s} = t^7 \left(\frac{2s}{s^2} \right) - \frac{4}{7} s^{-\frac{3}{7}} = \frac{2t^7}{s} - \frac{4}{7} s^{-\frac{3}{7}}$$

$$h_t(s, t) = \frac{\partial h}{\partial t} = 7t^6 \ln(s^2) - 27t^{-4}$$

Remember how to differentiate natural logarithms.

$$\frac{d}{dx}(\ln g(x)) = \frac{g'(x)}{g(x)}$$

(d) $f(x, y) = \cos\left(\frac{4}{x}\right) e^{x^2y-5y^3}$

Now, we can't forget the product rule with derivatives. The product rule will work the same way here as it does with functions of one variable. We will just need to be careful to remember which variable we are differentiating with respect to.

Let's start out by differentiating with respect to x . In this case both the cosine and the exponential contain x 's and so we've really got a product of two functions involving x 's and so we'll need to product rule this up. Here is the derivative with respect to x .

$$\begin{aligned} f_x(x, y) &= -\sin\left(\frac{4}{x}\right) \left(-\frac{4}{x^2}\right) e^{x^2y-5y^3} + \cos\left(\frac{4}{x}\right) e^{x^2y-5y^3} (2xy) \\ &= \frac{4}{x^2} \sin\left(\frac{4}{x}\right) e^{x^2y-5y^3} + 2xy \cos\left(\frac{4}{x}\right) e^{x^2y-5y^3} \end{aligned}$$

Do not forget the [chain rule](#) for functions of one variable. We will be looking at the chain rule for some more complicated expressions for multivariable functions in a later section. However, at this point we're treating all the y 's as constants and so the chain rule will continue to work as it did back in Calculus I.

Also, don't forget how to differentiate exponential functions,

$$\frac{d}{dx} \left(e^{f(x)} \right) = f'(x) e^{f(x)}$$

Now, let's differentiate with respect to y . In this case we don't have a product rule to worry about since the only place that the y shows up is in the exponential. Therefore, since x 's are considered to be constants for this derivative, the cosine in the front will also be thought of as a multiplicative constant. Here is the derivative with respect to y .

$$f_y(x, y) = (x^2 - 15y^2) \cos\left(\frac{4}{x}\right) e^{x^2y - 5y^3}$$

Example 2 Find all of the first order partial derivatives for the following functions.

(a) $z = \frac{9u}{u^2 + 5v}$

(b) $g(x, y, z) = \frac{x \sin(y)}{z^2}$

(c) $z = \sqrt{x^2 + \ln(5x - 3y^2)}$

Solution

(a) $z = \frac{9u}{u^2 + 5v}$

We also can't forget about the quotient rule. Since there isn't too much to this one, we will simply give the derivatives.

$$z_u = \frac{9(u^2 + 5v) - 9u(2u)}{(u^2 + 5v)^2} = \frac{-9u^2 + 45v}{(u^2 + 5v)^2}$$

$$z_v = \frac{(0)(u^2 + 5v) - 9u(5)}{(u^2 + 5v)^2} = \frac{-45u}{(u^2 + 5v)^2}$$

In the case of the derivative with respect to v recall that u 's are constant and so when we differentiate the numerator we will get zero!

(b) $g(x, y, z) = \frac{x \sin(y)}{z^2}$

Now, we do need to be careful however to not use the quotient rule when it doesn't need to be used. In this case we do have a quotient, however, since the x 's and y 's only appear in the numerator and the z 's only appear in the denominator this really isn't a quotient rule problem.

Let's do the derivatives with respect to x and y first. In both these cases the z 's are constants and so the denominator in this is a constant and so we don't really need to worry too much about it. Here are the derivatives for these two cases.

$$g_x(x, y, z) = \frac{\sin(y)}{z^2} \qquad g_y(x, y, z) = \frac{x \cos(y)}{z^2}$$

Now, in the case of differentiation with respect to z we can avoid the quotient rule with a quick rewrite of the function. Here is the rewrite as well as the derivative with respect to z .

$$g(x, y, z) = x \sin(y) z^{-2}$$

$$g_z(x, y, z) = -2x \sin(y) z^{-3} = -\frac{2x \sin(y)}{z^3}$$

We went ahead and put the derivative back into the "original" form just so we could say that we did. In practice you probably don't really need to do that.

(c) $z = \sqrt{x^2 + \ln(5x - 3y^2)}$

In this last part we are just going to do a somewhat messy chain rule problem. However, if you had a good background in [chain rule](#) this shouldn't be all that difficult of a problem. Here are the two derivatives,

$$\begin{aligned} z_x &= \frac{1}{2} \left(x^2 + \ln(5x - 3y^2) \right)^{-\frac{1}{2}} \frac{\partial}{\partial x} \left(x^2 + \ln(5x - 3y^2) \right) \\ &= \frac{1}{2} \left(x^2 + \ln(5x - 3y^2) \right)^{-\frac{1}{2}} \left(2x + \frac{5}{5x - 3y^2} \right) \\ &= \left(x + \frac{5}{2(5x - 3y^2)} \right) \left(x^2 + \ln(5x - 3y^2) \right)^{-\frac{1}{2}} \\ z_y &= \frac{1}{2} \left(x^2 + \ln(5x - 3y^2) \right)^{-\frac{1}{2}} \frac{\partial}{\partial y} \left(x^2 + \ln(5x - 3y^2) \right) \\ &= \frac{1}{2} \left(x^2 + \ln(5x - 3y^2) \right)^{-\frac{1}{2}} \left(\frac{-6y}{5x - 3y^2} \right) \\ &= -\frac{3y}{5x - 3y^2} \left(x^2 + \ln(5x - 3y^2) \right)^{-\frac{1}{2}} \end{aligned}$$

So, there are some examples of partial derivatives. Hopefully you will agree that as long as we can remember to treat the other variables as constants these work in exactly the same manner that derivatives of functions of one variable do. So, if you can do Calculus I derivatives you shouldn't have too much difficulty in doing basic partial derivatives.

There is one final topic that we need to take a quick look at in this section, implicit differentiation. Before getting into implicit differentiation for multiple variable functions let's first remember how implicit differentiation works for functions of one variable.

Example 3 Find $\frac{dy}{dx}$ for $3y^4 + x^7 = 5x$.

Solution

Remember that the key to this is to always think of y as a function of x , or $y = y(x)$ and so whenever we differentiate a term involving y 's with respect to x we will really need to use the chain rule which will mean that we will add on a $\frac{dy}{dx}$ to that term.

The first step is to differentiate both sides with respect to x .

$$12y^3 \frac{dy}{dx} + 7x^6 = 5$$

The final step is to solve for $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{5 - 7x^6}{12y^3}$$

Now, we did this problem because implicit differentiation works in exactly the same manner with functions of multiple variables. If we have a function in terms of three variables x , y , and z we will assume that z is in fact a function of x and y . In other words, $z = z(x, y)$. Then whenever we differentiate z 's with respect to x we will use the chain rule and add on a $\frac{\partial z}{\partial x}$. Likewise, whenever we differentiate z 's with respect to y we will add on a $\frac{\partial z}{\partial y}$.

Let's take a quick look at a couple of implicit differentiation problems.

Example 4 Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for each of the following functions.

(a) $x^3 z^2 - 5xy^5 z = x^2 + y^3$

(b) $x^2 \sin(2y - 5z) = 1 + y \cos(6zx)$

Solution

(a) $x^3 z^2 - 5xy^5 z = x^2 + y^3$

Let's start with finding $\frac{\partial z}{\partial x}$. We first will differentiate both sides with respect to x and remember to add on a $\frac{\partial z}{\partial x}$ whenever we differentiate a z from the chain rule.

$$3x^2z^2 + 2x^3z \frac{\partial z}{\partial x} - 5y^5z - 5xy^5 \frac{\partial z}{\partial x} = 2x$$

Remember that since we are assuming $z = z(x, y)$ then any product of x 's and z 's will be a product and so will need the product rule!

Now, solve for $\frac{\partial z}{\partial x}$.

$$\begin{aligned} (2x^3z - 5xy^5) \frac{\partial z}{\partial x} &= 2x - 3x^2z^2 + 5y^5z \\ \frac{\partial z}{\partial x} &= \frac{2x - 3x^2z^2 + 5y^5z}{2x^3z - 5xy^5} \end{aligned}$$

Now we'll do the same thing for $\frac{\partial z}{\partial y}$ except this time we'll need to remember to add on a $\frac{\partial z}{\partial y}$ whenever we differentiate a z from the chain rule.

$$\begin{aligned} 2x^3z \frac{\partial z}{\partial y} - 25xy^4z - 5xy^5 \frac{\partial z}{\partial y} &= 3y^2 \\ (2x^3z - 5xy^5) \frac{\partial z}{\partial y} &= 3y^2 + 25xy^4z \\ \frac{\partial z}{\partial y} &= \frac{3y^2 + 25xy^4z}{2x^3z - 5xy^5} \end{aligned}$$

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